

Filter_Function

Python's filter() is a built-in function used to select elements from an iterable based on a condition. It returns only those elements for which the given function returns True.

Syntax:

```
iterable

def func_name(n):
    condition...
    return True / False

res = filter(func_name, iterable)
print(list(res))
```

Example:-

```
my_list = [10, 15, 20, 25, 30]

def is_even(n):
    return n % 2 == 0

x = filter(is_even, my_list)
print(x)
print(list(x))

# O/P:--
# <filter object at 0x...>
# [10, 20, 30]
```

Example:-

```
my_tuple = (11, 22, 33, 44)

def greater_than_20(n):
    return n > 20

x = filter(greater_than_20, my_tuple)
print(x)
print(tuple(x))

# O/P:--
```

```
# <filter object at 0x...>  
# (22, 33, 44)
```

Example:

```
my_str = "Neeraj"  
  
def is_vowel(ch):  
    return ch.lower() in 'aeiou'  
  
x = filter(is_vowel, my_str)  
print(x)  
print(list(x))  
  
# O/P:-  
# <filter object at 0x...>  
# ['e', 'e', 'a']
```

Example:

```
my_list = [-10, 5, -3, 8, 0]  
  
def is_positive(n):  
    return n > 0  
  
x = filter(is_positive, my_list)  
print(list(x))  
  
# O/P:-  
# [5, 8]
```